

Chapter 6 Fluid resistance and energy loss

Section 1 Overview of flow resistance and energy loss

Section 2 Two regimes/states of fluid motion

Section 3 Laminar flow motion in circular tube

Section 4 Turbulent flow motion

Section 5 Frictional head loss of turbulent flow in
circular tubes

Section 6 Frictional head loss of turbulent flow in
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 Section 7 Minor head loss

Section 8 Concept of boundary layer

Section 9 Drag due to flow around a body

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Section 7 Local/minor head loss

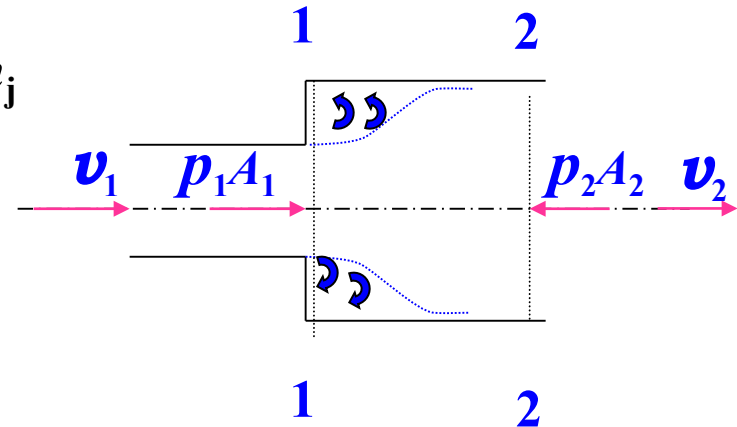
-  1. General expression for local head loss h_i
-  2. Determination of ζ for pipes with sudden enlargement
- 3. Determination of ζ for pipes with sudden contraction
- 4. Loss coefficient ζ at pipe inlet and outlet

Section 7 Local/minor head loss

1. General expression of local head loss h_j

$$h_j = \xi \frac{v^2}{2g}$$

where ξ ——local head loss coefficient



2. Determination of ζ for pipes with sudden enlargement

- Momentum equation (let $\beta_1 = \beta_2 = 1.0$; Horizontally placed pipe)

$$\sum F = p_1 A_1 - p_2 A_2 = \rho q_v (v_2 - v_1)$$

- Energy equation (let $\alpha_1 = \alpha_2 = 1.0$)

$$\frac{p_1}{\rho g} + \frac{v_1^2}{2g} = \frac{p_2}{\rho g} + \frac{v_2^2}{2g} + h_j$$

- Continuity equation

$$v_1 A_1 = v_2 A_2$$

Combining the above three equations gives:

$$h_j = \frac{(v_1 - v_2)^2}{2g}$$



$$h_j = \frac{(v_1 - v_2)^2}{2g}$$

Local head loss formula for pipes with sudden enlargement

$$\left. \begin{aligned} h_j &= \left(1 - \frac{A_1}{A_2}\right)^2 \frac{v_1^2}{2g} = \zeta \frac{v_1^2}{2g} \\ \zeta &= \left(1 - \frac{A_1}{A_2}\right)^2 \end{aligned} \right\} \quad \text{or} \quad \left. \begin{aligned} h_j &= \left(\frac{A_2}{A_1} - 1\right)^2 \frac{v_2^2}{2g} = \zeta \frac{v_2^2}{2g} \\ \zeta &= \left(\frac{A_2}{A_1} - 1\right)^2 \end{aligned} \right\}$$

Note:

- The local head loss of turbulence is proportional to **the square of velocity**.
- Attention to the relationship between local resistance coefficient and flow velocity.



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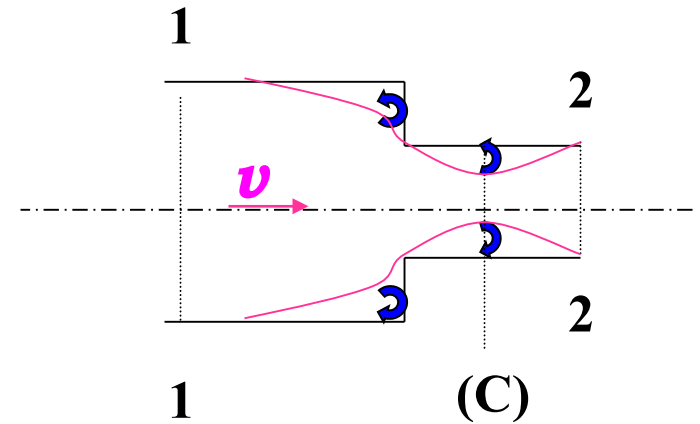
Section 7 Local/minor head loss

1. General expression for local head loss h_i
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-  3. Determination of ζ for pipes with sudden contraction
4. Loss coefficient ζ at pipe inlet and outlet

Loss coefficient ζ for pipes with sudden reduction contraction

$$h_j = \zeta \frac{v_2^2}{2g}$$

$$\zeta = 0.5\left(1 - \frac{A_2}{A_1}\right)$$



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$$h_j = \zeta \frac{v^2}{2g}$$

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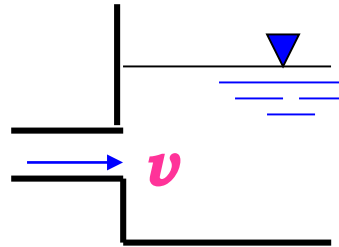
$$\zeta = \left(1 - \frac{A_1}{A_2}\right)^2$$

1. Pipe outlet

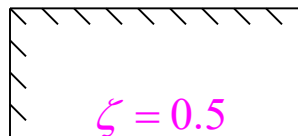
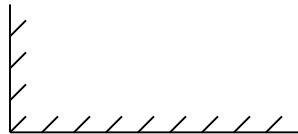
When pipe suddenly expands into a pool or container, $A_1 / A_2 \approx 0$

We have,

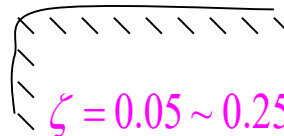
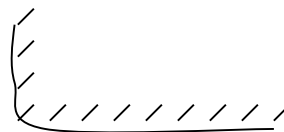
$$\zeta = 1.0$$



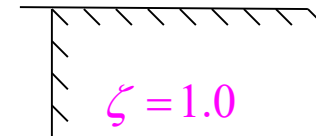
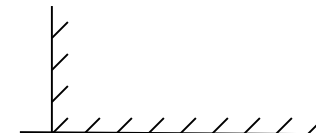
2. Pipe inlet



(a) Right-angle inlet
(直角进口)



(b) Rounded inlet
(圆角进口)



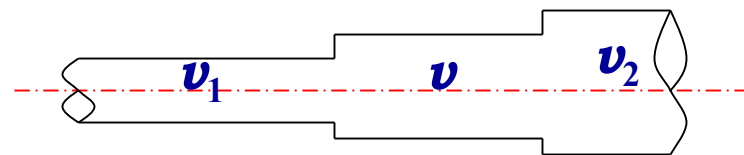
(c) Extended inlet
(外伸进口)

Example: In a pipe with sudden enlargement, the flow velocity changes from v_1 to v_2 . A medium-size pipe section is added in the middle to form two sudden pipe enlargements. The mutual interference of local resistance is omitted; that is, the superposition method can be used. Solve for:

(1) The flow velocity in the intermediate pipe, so that the total local head loss is minimal;

Solution: (1) The local head loss in the case with two sudden enlargements is

$$h_j = h_{j1} + h_{j2} = \frac{(v_1 - v)^2}{2g} + \frac{(v - v_2)^2}{2g}$$



To find the flow velocity v in the middle pipe that makes the total local head loss minimal, take the derivative such that:

$$\frac{dh_j}{dv} = 0$$

Therefore,
$$\frac{dh_j}{dv} = -\frac{2(v_1 - v)}{2g} + \frac{2(v - v_2)}{2g} = 0$$

which gives
$$v = \frac{v_1 + v_2}{2}$$

Example: In a pipe with sudden enlargement, the flow velocity changes from v_1 to v_2 . A medium-size pipe section is added in the middle to form two sudden pipe enlargements. The mutual interference of local resistance is omitted; that is, the superposition method can be used.

Solve for: (2) Calculate the total local head loss and compare it with that with only one enlargement.

Solution:(2) The total local loss is

$$h_j = \frac{(v_1 - \frac{v_1 + v_2}{2})^2}{2g} + \frac{(\frac{v_1 + v_2}{2} - v_2)^2}{2g} = \frac{(v_1 - v_2)^2}{4g}$$

The local head loss induced by one sudden enlargement is $h_j = \frac{(v_1 - v_2)^2}{2g}$.

Therefore, the local head loss in the case with two sudden enlargements is $1/2$ of that with one enlargement.



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- (1) In pipes with abrupt changes of diameter, when the flow direction changes while other conditions keep the same, whether the local head losses in the region with diameter change are equal in the cases of different flow directions? Why?

Unequal. Different alterations of solid boundaries (e.g. sudden enlargement, sudden contraction) cause different head loss.

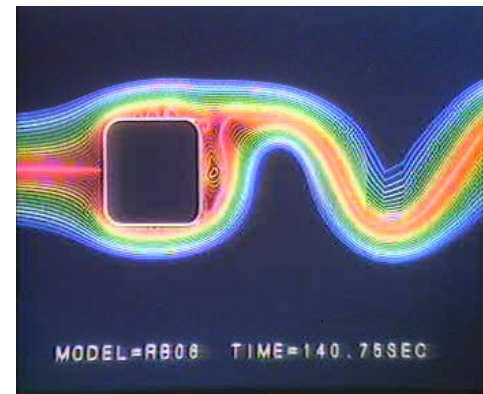
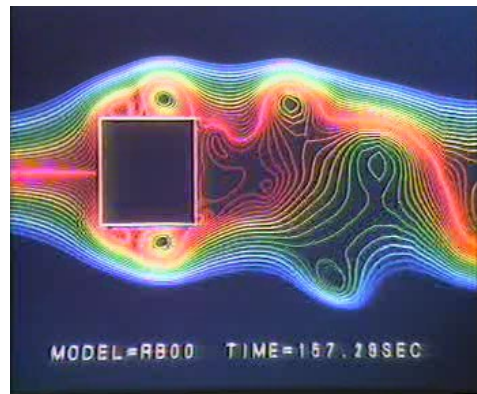
- (2) What factors affect the local resistance coefficient? What should be considered when selecting such a coefficient?

The sudden alteration of solid boundary, flow velocity;

The local resistance coefficient should correspond to flow velocity.

- (3) How to reduce the local head loss?

Make the solid boundary close to streamline.



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Section 8 Concept of boundary layer

- 👉 1. Boundary layer
- 2. Boundary layer separation

Section 8 Concept of boundary layer

1. Boundary layer

- **Definition** (亦称附面层) : In the case of large Re , a fluid with small viscosity (e.g. air or water) flows along a solid wall (or solid moves in fluid) and the fluid motion with a thin layer adjacent to the wall is significantly affected by viscosity. This layer is called the boundary layer.
- **The solution of flow field can be divided into two zones**
 - (1) **Flow in boundary layer**: Viscous effect of fluid must be taken into account. An approximate solution can be obtained by using momentum equation.
 - (2) **Flow out of boundary layer**: Treated as an ideal fluid flow and can be solved as potential flow.



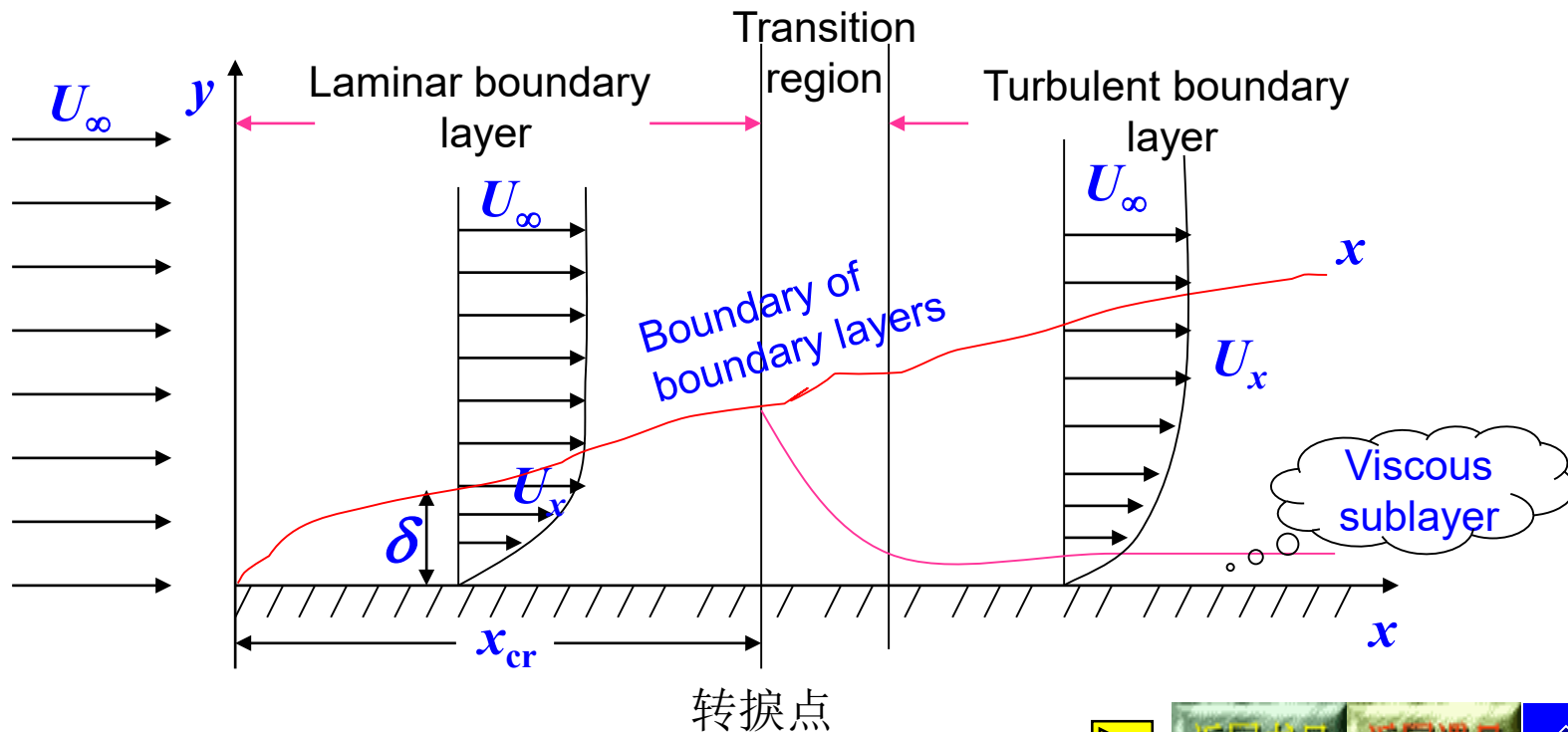
• Description of boundary layer

For the thin layer attaching to solid boundary, in which the shear stress is large and viscous effect cannot be ignored, Prandtl called it **boundary layer**.

Two flow regimes in boundary layer:
laminar flow and turbulent flow.

Re at the transition location is
the critical Reynolds number:

$$Re_{cr} = U_{\infty} x_{cr} / \nu$$



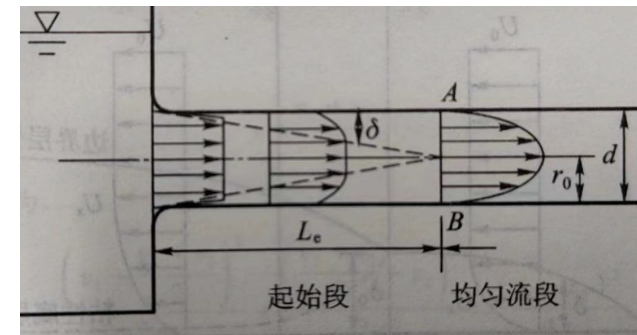
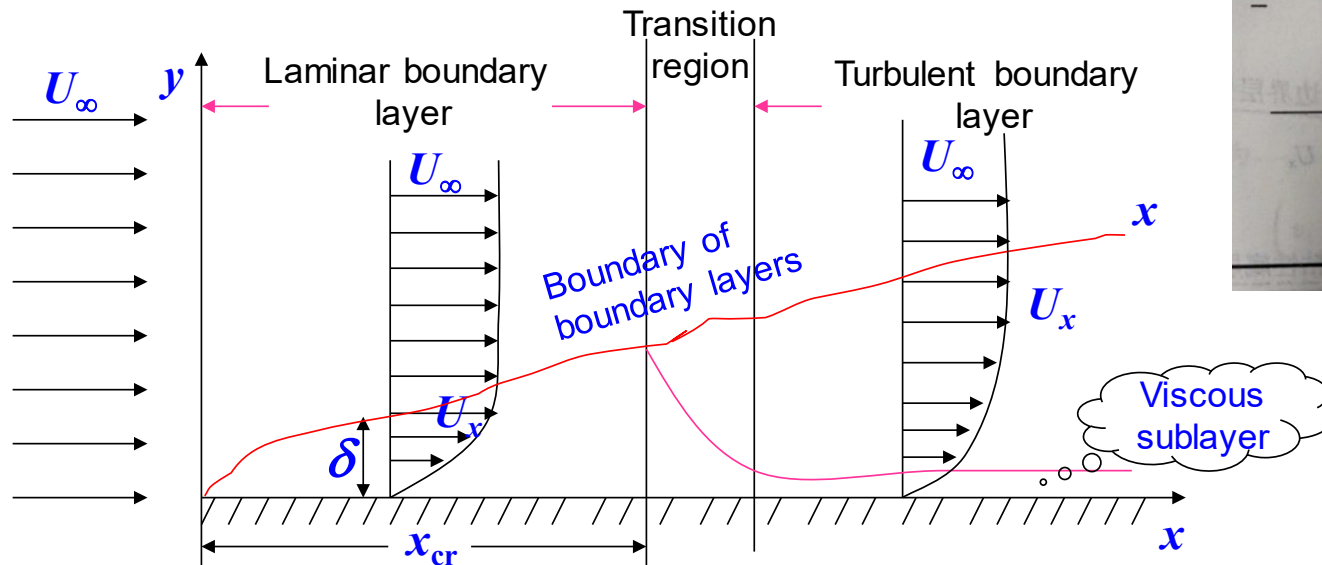
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• Laminar and turbulent boundary layer

- **Laminar boundary layer:** When the boundary layer thickness δ is small, the velocity gradient in the layer is large, and the effect of viscous stress is also large. In this condition, the flow in the boundary layer is laminar flow, called the laminar boundary layer.
- **Turbulent boundary layer:** When Re reaches a certain value, the laminar flow in the boundary layer evolves into turbulence after passing through a transition zone, becoming the turbulent boundary layer.
- In turbulent boundary layer, du_x/dy is still very large closest to the boundary, and viscous force plays a dominant role. The flow pattern is still laminar. So, there is a **viscous sublayer** in the turbulent boundary layer.



- **Smooth-plate boundary layer (光滑平板边界层)**

(1) Range of critical Reynolds number:

$$3 \times 10^5 < Re_{cr} = Ux_{cr} / \nu < 3 \times 10^6$$

The **critical Reynolds number for boundary layer** is not a constant, but **related to the disturbance intensity of the incoming flow**. If the incoming flow is disturbed with strong pulsation, the transition of flow regime can occur at a relatively low Reynolds number.

(2) Boundary layer thickness

Laminar boundary layer $\delta = \frac{5x}{Re_x^{1/2}}$

Turbulent boundary layer $\delta = \frac{0.377x}{Re_x^{1/5}}$



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1. Boundary layer

 2. Boundary layer separation

2. Separation of boundary layer

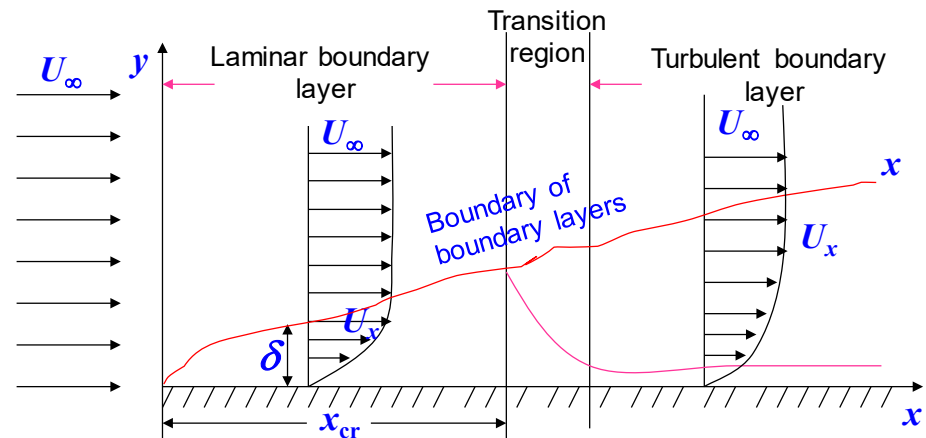
- **Definition:** Due to the pressure increases along the flow direction, the phenomenon of fluid leaving the wall in the boundary layer is called the boundary layer separation (边界层与过流壁面脱离的现象) .



- **Boundary layer separation of flow around flat plate**

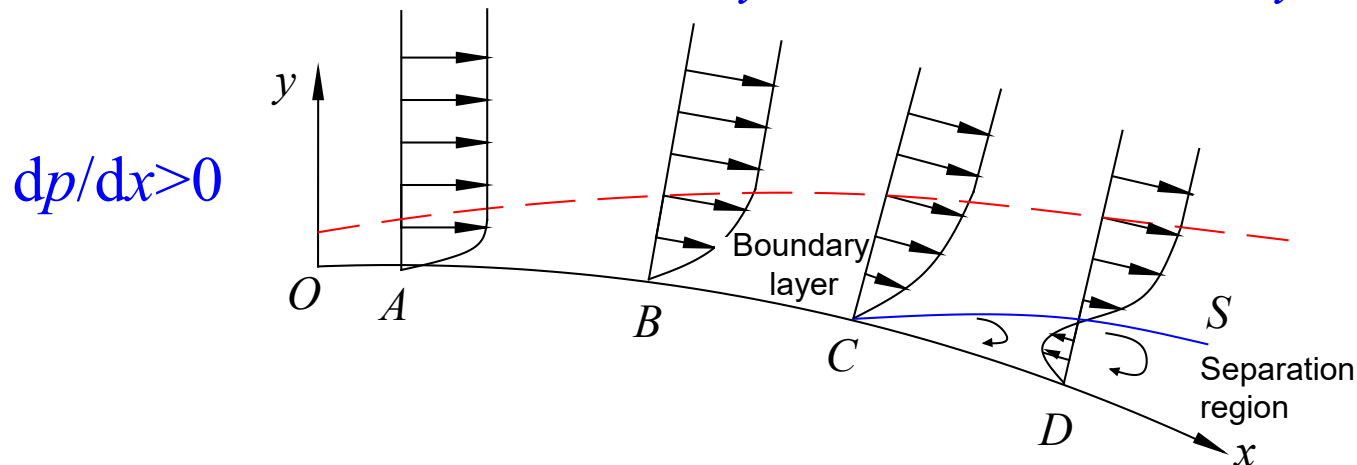
(1) If the pressure gradient remains zero, i.e., $dp/dx=0$

No matter how long the plate is, **no separation will occur**, and the boundary layer will thicken continuously along the flow direction.



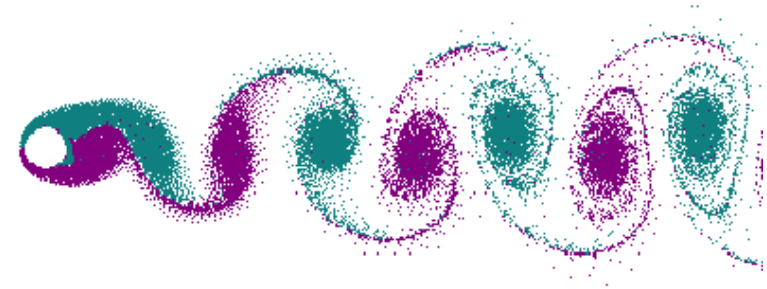
(2) If pressure increases along flow direction, i.e., $p_2 > p_1$ or $dp/dx > 0$

- Boundary layer thickens rapidly.
 - The increase of pressure (decrease of flow velocity) and the increase of resistance make the drop of momentum in the boundary layer.
 - If the two factors act together at a distance long enough, the fluid flow in the boundary layer can stagnate, and the separation occurs.
 - From the separation point C, the boundary streamline leaves the boundary, and a backflow (or vortex) is formed downstream near the wall.
- At the separation point : $(\frac{\partial u}{\partial y})_{y=0} = 0$ $\tau_0 = \mu(\frac{\partial u}{\partial y})_{y=0} = 0$



• Wake

Wake: The downstream area surrounded by separation streamlines and solid boundaries.

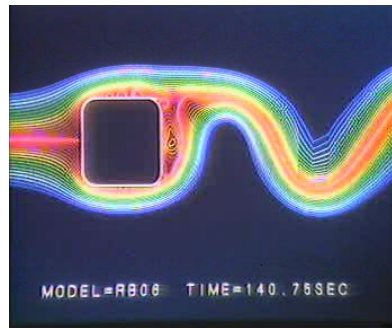
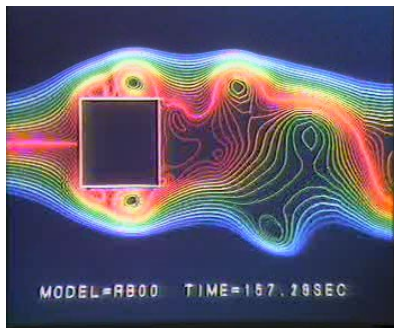


Karman vortex street

Adverse effects of wake:

- Increase local flow resistance, reduce effective energy of fluid flows
- Backflows/vortexes cause scouring or sedimentation
- Vortexes cause structural vibrations

Main ways to mitigate wake: Make the shape of solid bodies streamlined (流线型) as much as possible.



1. Whether the flows in the boundary layer must be laminar flow? What are the primary factors affecting the flow regime in boundary layer?

No, they can be laminar and turbulent boundary layer.

Primary factors include viscosity, velocity, distance

2. How is the boundary layer separation formed? How to reduce the wake region ?

- Due to the increase of pressure along the flow direction and the existence of resistance, the momentum in the boundary layer decreases, forming the separation of boundary layer.
- The wake region can be reduced by making the flow around the fluid as streamlined as possible.

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